Skills Summary

Square or Cubic? Getting the Unit Right

Use this reference while completing your worksheet or when studying.

Skill 1 — Read the unit off the formula

Rule: count the lengths you multiplied.

- one length (perimeter, an edge) → plain units: cm, m, in
- **two** lengths (area, surface area) → **square** units
- **three** lengths (volume) → **cubic** units

Area of a rectangle $= \ell \times w$. Volume of a box $= \ell \times w \times h$. Surface area of a cube $= 6 \times (\text{edge} \times \text{edge})$.

Example: A box is 5 centimeters long, 3 centimeters wide and 2 centimeters tall. Find the volume, with its unit.
Step 1. Volume fills a space, so I pack unit cubes in — that means multiplying all three edges, and three lengths always give cubic units.
Step 2. One layer on the floor holds $5 \times 3 = 15$ cubes, and there are 2 layers: $15 \times 2 = 30$.
$\Rightarrow$ volume = **30** cubic centimeters

Try it: A box is 6 centimeters long, 2 centimeters wide and 4 centimeters tall. Find its volume in cubic centimeters.

check: **48**

Skill 2 — Find and fix a squared-for-cubed mistake

How to catch it: check the *number* and the *label* separately.

Count how many lengths were actually multiplied. If only two were multiplied but the label says cubic, the number is an area wearing the wrong coat — and the fix is to multiply by the missing third length, not to change the label.

Cube shortcut: face area = edge × edge; volume = edge × edge × edge.

Example: A student found the volume of a cube with edges 5 centimeters long and wrote 25 cubic centimeters. Find and fix the mistake.
Step 1. Only two edges were multiplied, so 25 measures one flat face — an area number with a cubic label, which is the classic squared-for-cubed slip.
Step 2. The cube is 5 layers deep, so the missing third edge goes in: $25 \times 5 = 125$. $\Rightarrow$ volume = **125** cubic centimeters

Try it: A cube has edges 3 centimeters long. Find its volume in cubic centimeters.

check: **27**

Skill 3 — Convert square and cubic units

Rule: the length factor is used once per direction.

If the lengths convert with a factor f, then square units convert with $f \times f$ and cubic units convert with $f \times f \times f$. Never reuse the plain length factor on an area or a volume.

Example: How many square centimeters cover one square meter? (One meter is 100 centimeters.)

Step 1. A square meter is a square with a meter on each side, and a square has two directions — so the factor 100 is used twice.

Step 2. The square holds 100 rows of 100 small squares: $100 \times 100 = 10000$. $\Rightarrow$ **10000** square centimeters in one square meter

Try it: How many cubic inches fill one cubic foot? (One foot is 12 inches.)

check: **1728**

(!) Watch out: The unit is not decoration you add at the end — it is a record of how many lengths you multiplied. If your number came from two lengths, no label can make it a volume; go back and find the third length.